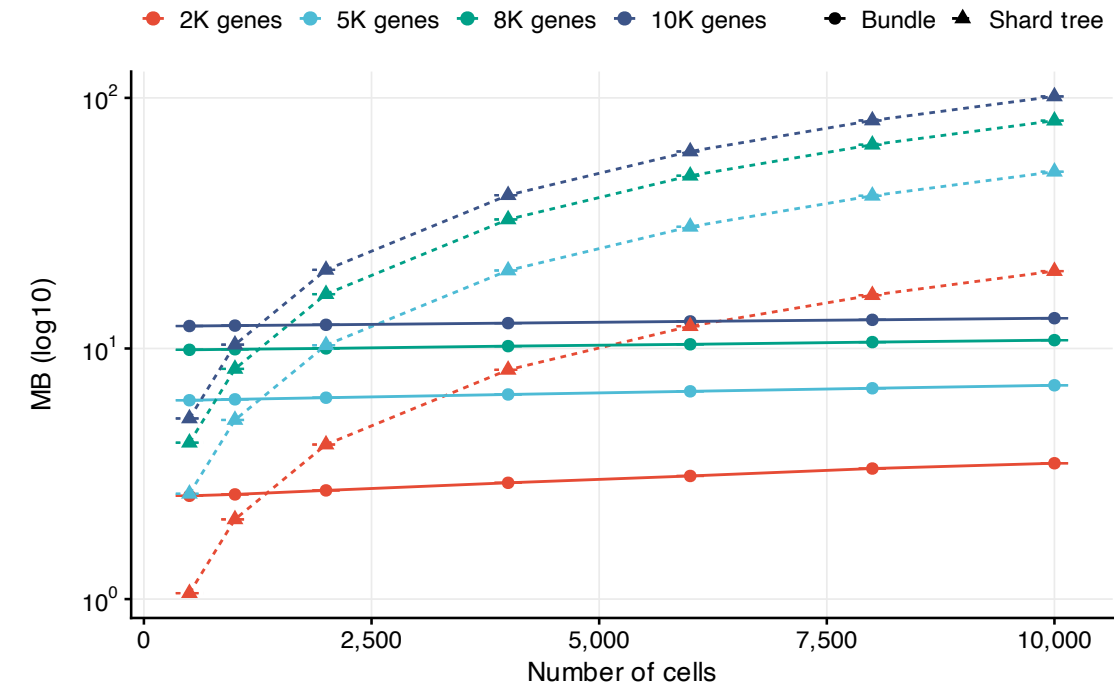
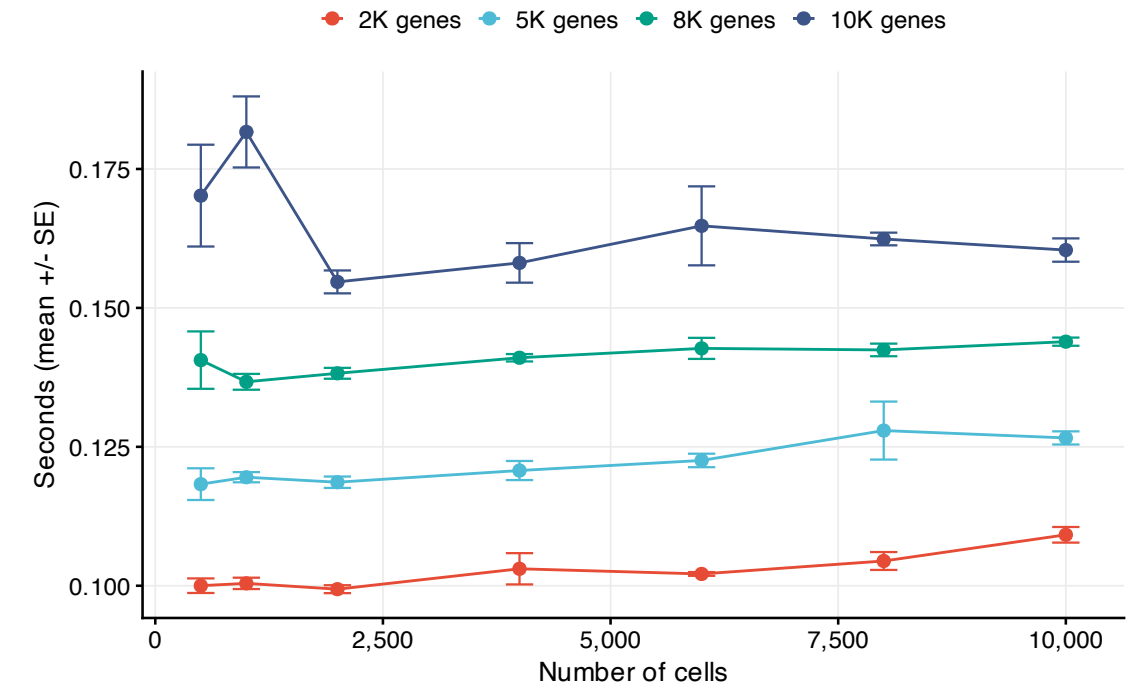
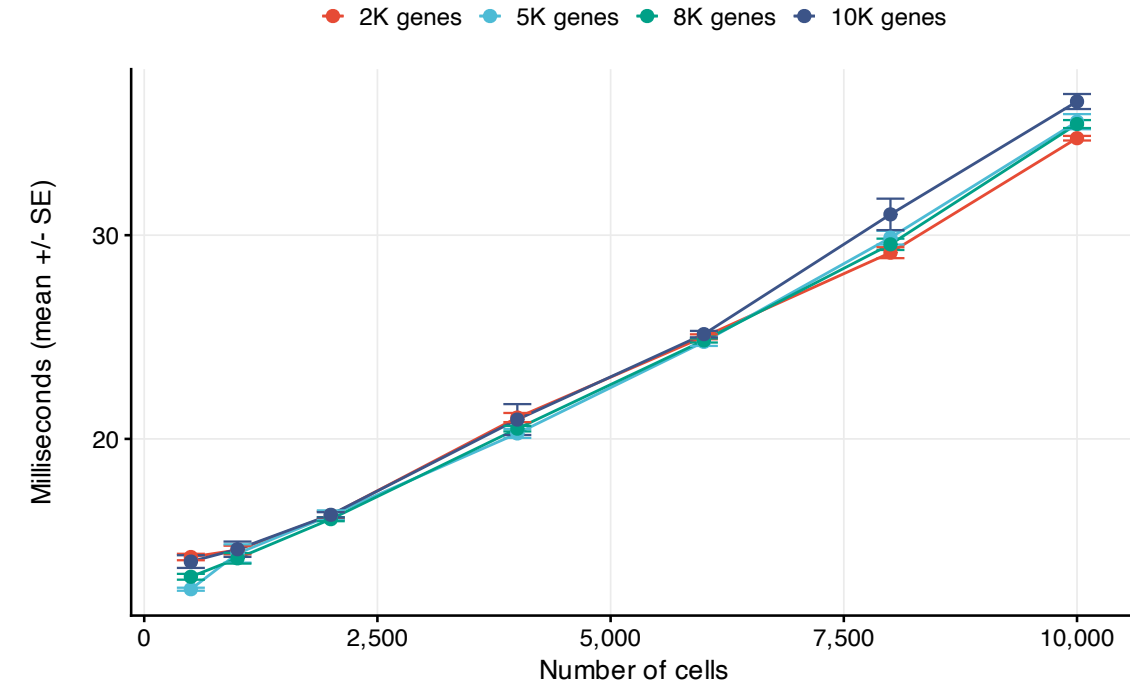
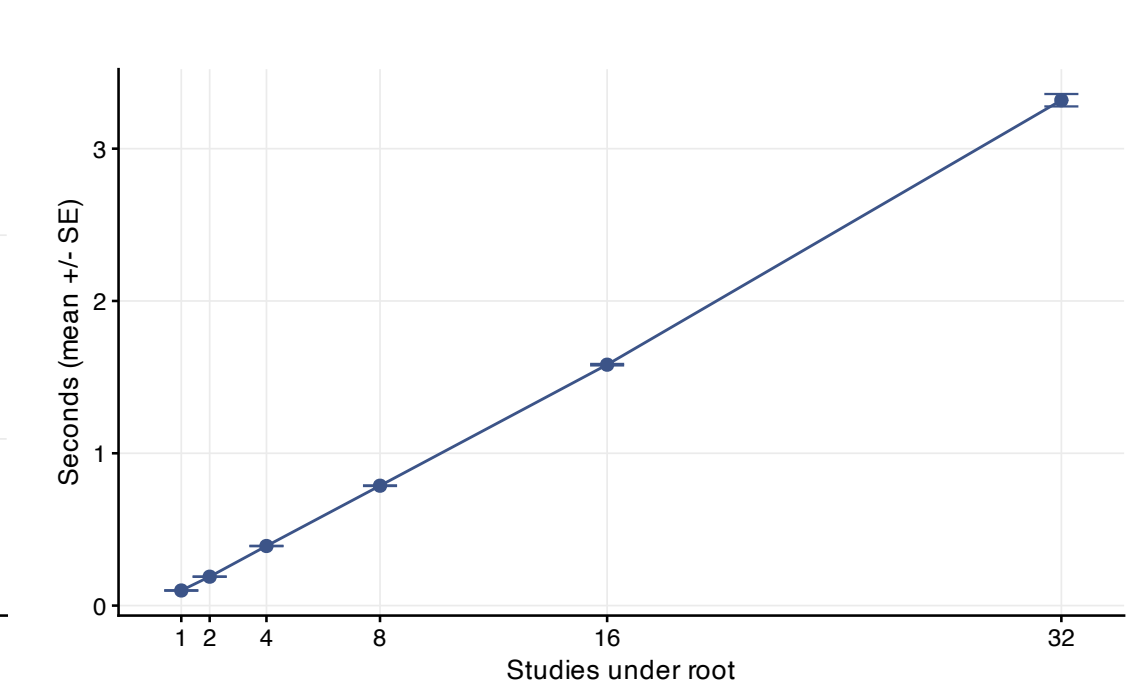
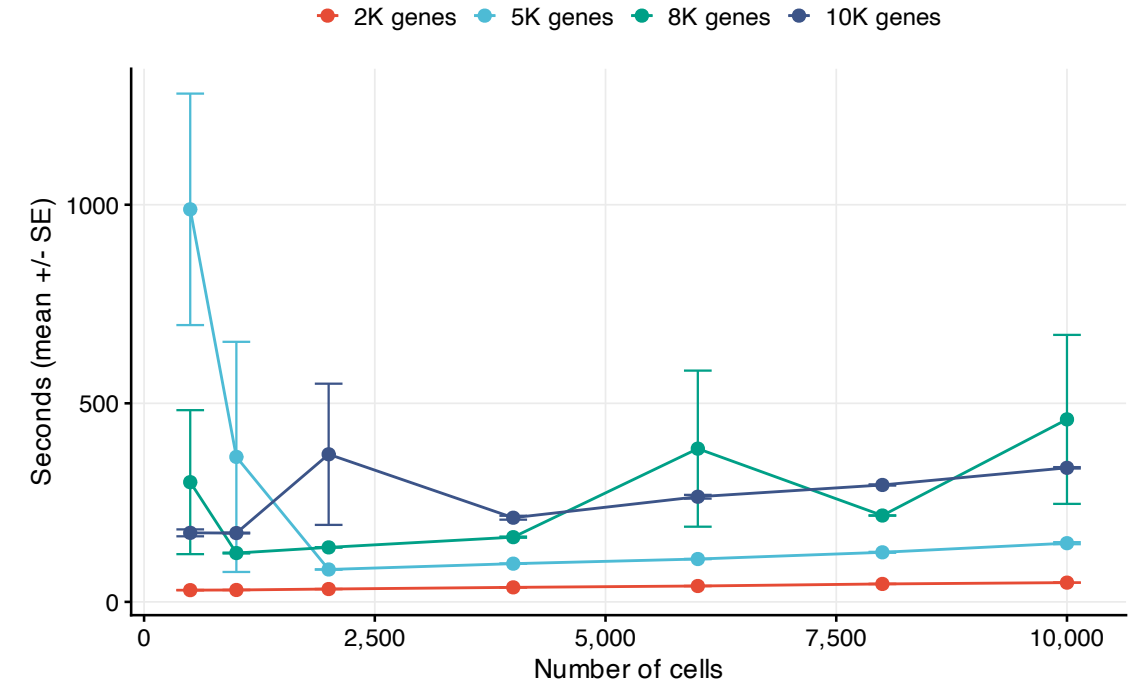


A**Bundle size vs shard-tree size****B****load_study() cold-start latency****C****gene_values() median fetch latency****D****discover_studies() scaling****E****prepare_study_data() wall time****F****prepare_study_data() peak memory**